HQDFM Stack-up Impedance Report

Layer Count: 8

Stack-up Diagram:

L1	完成铜厚1oz(0.33oz+电镀铜)	铜箔	1	0.0115mm	Residual Copper Rate:70
	2116	pp	2116	0.1250mm	
L2/L3	0.3 含铜 1/1	芯板	含铜	0.3000mm	Residual Copper Rate:60
	2313	pp	2313	0.1030mm	
	2313	pp	2313	0.1030mm	
L4/L5	0.3 含铜 1/1	芯板	含铜	0.3000mm	Residual Copper Rate:50
	2313	pp	2313	0.1030mm	
	2313	pp	2313	0.1030mm	
L6/L7	0.3 含铜 1/1	芯板	含铜	0.3000mm	Residual Copper Rate:50
	2116	pp	2116	0.1250mm	
L8	完成铜厚1oz(0.33oz+电镀铜)	铜箔	1	0.0115mm	Residual Copper Rate:70

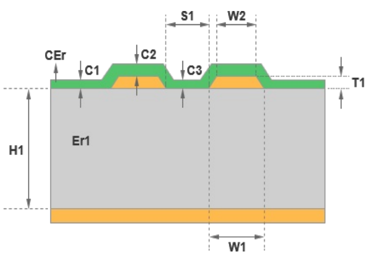
Board Thickness

Orig. Final Board Thickness:	1.6000mm +/-10%
Laminated:	1.4954 mm

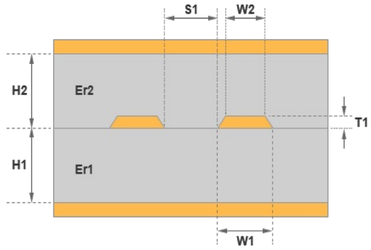
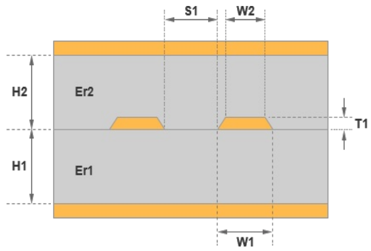
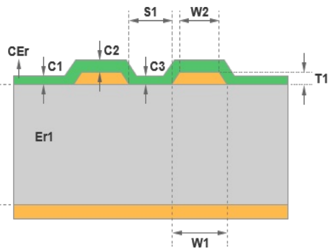
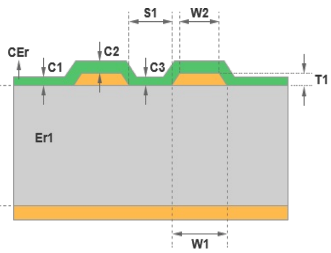
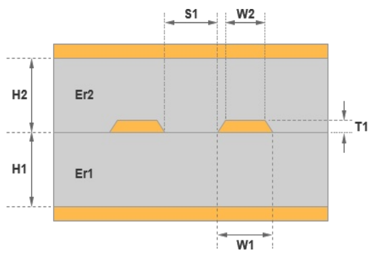
Impedance Controlled Traces (Lengths in mil)

No.	Type	Layer	Orig. Trace Width	Orig. Trace Spacing	Orig. Distance To Copper	Upper Ref	Lower Ref	Trace Width	Trace Spacing	Distance To Copper	Target (ohm)	Tol.	Actual (ohm)
1	Outer Differentia	L1	5.44	5.00	/	/	L2	5.80	4.64	/	90.00	(+/-10%)	89.60
2	Inner Differentia	L3	6.69	6.30	/	L2	L4	5.80	7.19	/	90.00	(+/-10%)	90.02
3	Inner Differentia	L6	6.69	6.30	/	L5	L7	5.80	7.19	/	90.00	(+/-10%)	90.02
4	Outer Differentia	L1	5.01	7.09	/	/	L2	5.30	6.80	/	100.00	(+/-10%)	100.45
5	Outer Differentia	L8	5.01	7.09	/	L7	/	5.30	6.80	/	100.00	(+/-10%)	100.45
6	Inner Differentia	L3	5.00	10.24	/	L2	L4	5.20	10.04	/	100.00	(+/-10%)	99.68
7	Inner Differentia	L6	5.00	10.24	/	L5	L7	5.20	10.04	/	100.00	(+/-10%)	99.68
8	Outer Differentia	L8	5.44	5.00	/	L7	/	5.80	4.64	/	90.00	(+/-10%)	89.60

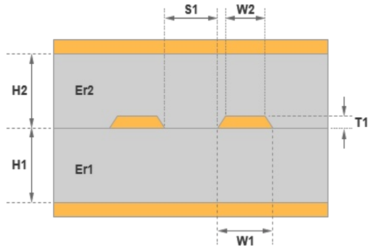
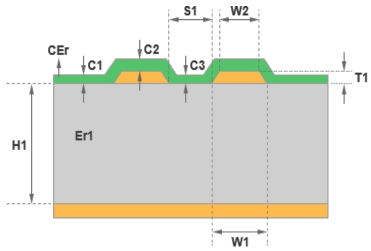
Impedance Calculations

Model	Values	Impedance Information
	W1 : 5.80 W2 : 5.00 S1 : 4.64 H1 : 4.4173 Er1 : 4.2000 T1 : 1.3780 C1 : 1.1969 C2 : 0.5000 C3 : 1.1969 CEr : 3.5000	Outer Differential Controlled Layer: L1 Reference: &L Lower Trace Width:5.80mil Trace Separation:4.64mil Required Impedance: 90.00 (+/-10%) Actual Impedance: 89.60 ohm

Impedance Calculations

Model	Values	Impedance Information
	W1 : 5.80 W2 : 5.11 S1 : 7.19 H1 : 9.2913 H2 : 8.1102 Er1 : 4.2000 Er2 : 4.2000 T1 : 1.2598	Inner Differential Controlled Layer: L3 Reference: L2 Lower Trace Width:5.80mil Trace Separation:7.19mil Required Impedance: 90.00 (+/-10%) Actual Impedance: 90.02 ohm
	W1 : 5.80 W2 : 5.11 S1 : 7.19 H1 : 9.2913 H2 : 8.1102 Er1 : 4.2000 Er2 : 4.2000 T1 : 1.2598	Inner Differential Controlled Layer: L6 Reference: L5 Lower Trace Width:5.80mil Trace Separation:7.19mil Required Impedance: 90.00 (+/-10%) Actual Impedance: 90.02 ohm
	W1 : 5.30 W2 : 4.50 S1 : 6.80 H1 : 4.4173 Er1 : 4.2000 T1 : 1.3780 C1 : 1.1969 C2 : 0.5000 C3 : 1.1969 CEr : 3.5000	Outer Differential Controlled Layer: L1 Reference: &L Lower Trace Width:5.30mil Trace Separation:6.80mil Required Impedance: 100.00 (+/-10% Actual Impedance: 100.45 ohm
	W1 : 5.30 W2 : 4.50 S1 : 6.80 H1 : 4.4173 Er1 : 4.2000 T1 : 1.3780 C1 : 1.1969 C2 : 0.5000 C3 : 1.1969 CEr : 3.5000	Outer Differential Controlled Layer: L8 Reference: L7 Lower Trace Width:5.30mil Trace Separation:6.80mil Required Impedance: 100.00 (+/-10% Actual Impedance: 100.45 ohm
	W1 : 5.20 W2 : 4.51 S1 : 10.04 H1 : 9.2913 H2 : 8.1102 Er1 : 4.2000 Er2 : 4.2000 T1 : 1.2598	Inner Differential Controlled Layer: L3 Reference: L2 Lower Trace Width:5.20mil Trace Separation:10.04mil Required Impedance: 100.00 (+/-10% Actual Impedance: 99.68 ohm

Impedance Calculations

Model	Values	Impedance Information
	W1 : 5.20 W2 : 4.51 S1 : 10.04 H1 : 9.2913 H2 : 8.1102 Er1 : 4.2000 Er2 : 4.2000 T1 : 1.2598	Inner Differential Controlled Layer: L6 Reference: L5 Lower Trace Width:5.20mil Trace Separation:10.04mil Required Impedance: 100.00 (+/-10% Actual Impedance: 99.68 ohm
	W1 : 5.80 W2 : 5.00 S1 : 4.64 H1 : 4.4173 Er1 : 4.2000 T1 : 1.3780 C1 : 1.1969 C2 : 0.5000 C3 : 1.1969 CEr : 3.5000	Outer Differential Controlled Layer: L8 Reference: L7 Lower Trace Width:5.80mil Trace Separation:4.64mil Required Impedance: 90.00 (+/-10%) Actual Impedance: 89.60 ohm